AURAREFLEX

Real-Time Visual (LED) & Auditory (Buzzer) Dual-Participant Reaction Time Measurement System

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01 ABSTRACT & MOTIVATION

AuraReflex measures reaction time for two participants simultaneously. An ESP32 microcontroller presents an LED or buzzer stimulus, timestamps each response with microsecond precision using digital noise filters, and streams data over UART to a Python desktop application for CSV logging and 4-panel analysis.

02 SYSTEM ARCHITECTURE & DATA FLOW

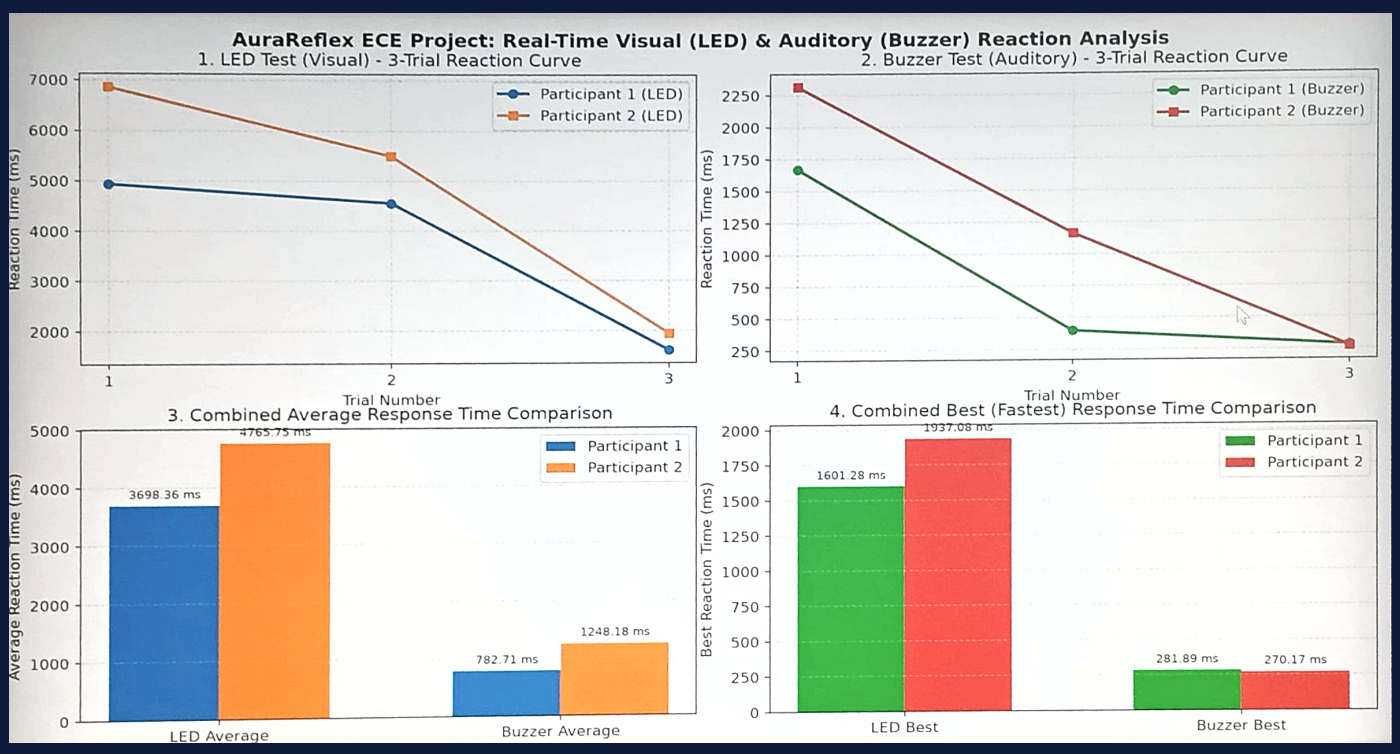
Stimulus Layer
ESP32 drives Red LED (visual) or 2-pin Buzzer (auditory).

Participant Input
Tactile push buttons wired on GPIO 32 & 33.

ESP32 Controller
C++ / PlatformIO microsecond delta (Δt) timing.

PC Analytics
Python & Matplotlib live plots + automated CSV export.

03 EXPERIMENTAL RESULTS & PERFORMANCE



AUDITORY EDGE
3.8–4.7× faster

PROJECT RECORD · P2 BUZZER
270.17 ms

LED AVERAGES
P1 3698.36 ms
P2 4765.75 ms

BUZZER AVERAGES
P1 782.71 ms
P2 1248.18 ms

LEARNING CURVE
Improved T1 → T3

04 ENGINEERING & PRACTICAL APPLICATIONS

Driver Fatigue Screening
Assessing cognitive latency before driving.

Sports Reflex Training
Tracking athletic reflex improvements trial-by-trial.

Neurological Baselines
Low-cost portable tool for reflex health assessment.

E-Sports Benchmarking
Quantifying human audio vs. visual reaction differentials.

Industrial Safety & HMI
Evaluating operator response speeds to optical vs. acoustic alarms.

05 REFERENCES

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